	RV College of Engineering® **Department of Computer Science and Engineering** **CIE - I: Test Paper**		
Course & Code	**Deep Learning** **(**CS374TFA**)**		**Semester: VII**
Date: 07/11/2025	**Duration:**90 minutes	**Max.Marks** : 50 Marks	**Staff : RJ**
USN :	**Name :**		

NOTE: ***Answer all the questions***

Part -A

Sl.no.	Questions	Marks	BT	CO
1.	Define convolution and write its mathematical expression.	2	L2	CO2
2.	What is the main limitation of a single-layer feedforward network?	1	L2	CO2
3.	What property of convolution makes it useful for feature extraction in CNNs?	2	L1	CO2
4.	Given the following **feature map**: $\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 1 & 2 \\ 3 & 1 & 2 & 0 \\ 1 & 2 & 3 & 4 \end{bmatrix}$ Apply 2X2 Average pooling with stride=2.Compute the pooled output feature map.	2	L3	CO2
5.	State the two main characteristics of knowledge representation.	2	L2	CO1
6.	What are the two main categories of learning processes in neural networks?	1	L2	CO1

Part B

Sl.no.	Questions	Marks	BT	CO
1.	What is a neuron? With a neat diagram, briefly explain the basic elements of a neuron.	10	L2	CO1
2.	With a neat diagram, briefly explain the different classes of neural network architectures	10	L2	CO1

3.	Compare max pooling, average pooling, *and* L2 pooling, and analyze how each affects the learned feature representations and invariance properties of the network.	10	L4	CO3
4.	A neuron has three inputs: $x_1 = 0.5, x_2 = -1.0, x_3 = 2.0$ with the corresponding weights $w_1 = 0.8, w_2 = -0.4, w_3 = 0.3$, and bias $b = 0.2$. Compute the output of the neuron for the following activation functions: i) Threshold (Heaviside) function ii) Sigmoid function iii) Hyperbolic Tangent Function Show all intermediate steps, including the calculation of the induced local field, and final outputs.	10	L3	CO4
5.	A 3 × 3 image patch I and a 2 × 2 kernel K are given below: $I = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 1 \\ 2 & 3 & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ a) Perform a **valid 2-D convolution** (no padding, stride = 1) and compute the full 2 × 2 output feature map. b) If the same kernel were applied as a **cross-correlation**, show which entries would differ. c) State one practical reason why kernel flipping is rarely used in CNN implementations.	10	L3	CO2

COURSE OUTCOMES:

CO1:	Explore key theoretical concepts like the Universal Approximation Theorem, vanishing / exploding gradients, and optimization methods.
CO2:	Analyse the fundamental concepts of Deep Learning, and its various architecture learning models, including Neural Networks, backpropagation, gradient descent, and different Network Architectures (feedforward, convolutional, recurrent) Learning tasks for various applications.
CO3:	Apply the Deep learning model approaches to know the strengths and weaknesses of the architecture by empirical results. Apply appropriate concepts like Recurrent, Recursive Nets and Auto-encoder models to specific real time projects and analyse the Optimization techniques.
CO4:	Designing and implement a Deep Learning model as part of an experiential learning initiative in teams to solve societal and environmental problems. Ability to fine tune the model parameters to improve performance, explore and understand the ethical implications and societal impact of deploying deep learning systems in real-world scenarios.

	L1	**L2**	**L3**	**L4**	**L5**	**L6**	**CO1**	**CO2**	**CO3**	**CO4**
Marks	2	26	22	10	-	-	13	17	10	10